

Chapter IV

VIRAL AGENTS CAUSING GASTROENTERITIS

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- ☐ Rotavirus
- ☐ Enteric adenoviruses
- ☐ Calicivirus
- ☐ Astrovirus
- ☐ Norwalkviruses
- ☐ Enteroviruses of Picornaviridae

Viruses as cause of Diarrhea

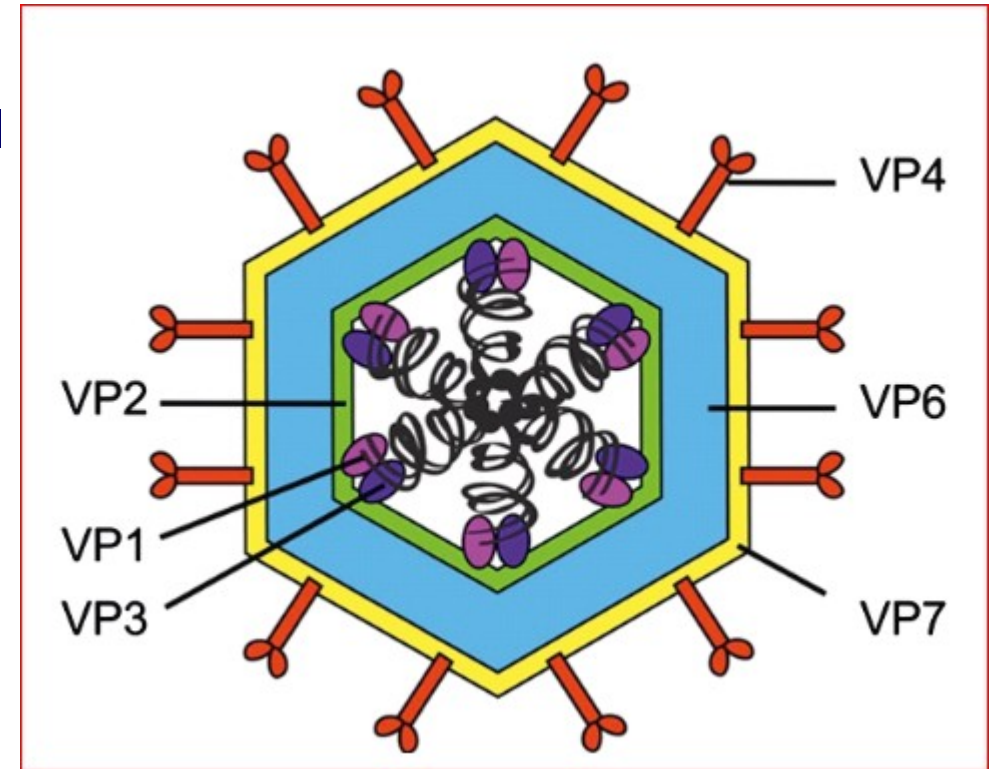
- Highly communicable - spreads rapidly
- Viruses large numbers in stools
- Significant titer of antibodies
- Reproduce disease
- Exclude other causes
- Rapid onset → recovery- mild systemic signs

ROTAVIRUS

- ds RNA, linear, segmented (11 segments); member of *Reoviridae*
- Classified on the basis of antigens on viral proteins VP 6 → 5 groups (A-E).

Humans (A,B,C) A most important

- Icosahedral, posses **three concentric capsid shell**
- No envelope
- Nine structural proteins, core- several enzymes
- Cytoplasmic replication, Inclusions in cytoplasm
- Genetic reassortment common
- Stable to heat, pH 3-9, , ethanol, phenol, chlorine



ROTAVIRUS

- Rotaviruses genome- 11 segments of double-stranded RNA that encodes
 - 6 structural (VP) and 5 or 6 nonstructural (NSP) proteins .
- VP 6- subgroup antigen
- Outer VP4 and VP7 determine serotype specificity, induce neutralizing antibodies
- Account for 40 to 60% of cases of acute gastroenteritis
- Virus is stable in the environment
- Relatively resistant to handwashing agents
- Susceptible to disinfection with 95% ethanol, 'Lysol', formalin

Pathogenesis

- Destroys villus cells of jejunum and duodenum
- Absorptive surface is decreased
- Enterotoxin-like effects are also present

Manifestations

- Short incubation period, vomiting, and watery diarrhea can lead to dehydration

Diagnosis

- Antigen detection in stool by ELISA, LA (for Group A rotavirus)
- EM- non-Group A viruses also
- Culture- Group A rotaviruses can be cultured in monkey kidney cells
- Serology for epidemiologic studies

Treatment and prevention

- Live, attenuated, or recombinant vaccines are feasible
- Rigorous hygienic measures

HUMAN CALICIVIRUSES

- Family *CALCIVIRIDAE*
- Non-enveloped RNA viruses with ss RNA, linear + sensed, non segmented
- Contain a single capsid protein
- **Replication:** Cytoplasm
- Norwalk viruses **most significant member** major cause of **nonbacterial epidemic gastroenteritis**
- Non-cultivable
- Two genera that cause diarrhea are recognized within this family
 - Norwalk-like and
 - Sapporo-like

- Epidemic nonbacterial gastroenteritis is characterized by
 - absence of bacterial pathogens;
 - gastroenteritis with **rapid onset** and recovery and relatively **mild systemic signs**;
 - an epidemiologic pattern of a highly communicable disease that spreads rapidly with no particular predilection in terms of age or geography.
- Their infectivity persists after exposure to acid, ether, and heat.
- No treatment or vaccine exists
- Prevention requires **good hygienic measures**.
- PCR is the most widely used technique
- Characteristics of Norwalk virus include a **low infectious dose** (as few as 10 virus particles)
- Fecal-oral spread is probably the **primary means** of transmission of Norwalk virus.

ASTROVIRUS

- Small + sense, ss, non segmented RNA virus 27-30 nm, family *Astroviridae*
- Non-enveloped
- EM appearance of a 5 or 6 **pointed star** *within* smooth edge
- At least 8 serotypes of human viruses
- Astroviruses cause diarrheal illness and may be shed in extraordinarily large quantities in feces.
- transmitted by the fecal-oral route through contaminated food or water, person-to-person contact, or contaminated surfaces
- Detected by EM

Human Enteroviruses

- Belong to Picornaviridae
- Transient inhabitants of human alimentary tract
- **At least 72 serotypes are known: divided into 5 groups**
 - Polioviruses
 - Coxsackieviruses A viruses
 - Coxsackie B viruses
 - Echoviruses
 - **Enteroviruses**
- Very resistant to environment conditions, GI low pH-fecal oral route

- Single stranded naked RNA virus with icosahedral symmetry
- Capsid has 4 proteins **Polypeptide**, VP1, VP2, VP3 and VP4 arranged with icosahedral symmetry around a positive sense genome.
- **Transmission involves**
 - Fecal – oral route: poor hygiene, dirty diapers(especially in day-care settings)
 - Ingestion via contaminated food and water
 - Contact with infected hands
 - Inhalation of infectious aerosols

Arthropod-Borne and Other Zoonotic Viruses

- The zoonotic viruses comprise more than 400 agents
- Members of the group have their ultimate reservoirs in **lower vertebrates or insects.**
- They are from diverse taxonomic families of RNA viruses that primarily include:
 - Togaviruses,
 - Bunyaviruses,
 - Reoviruses,
 - Arenaviruses, and
 - Filoviruses

Arboviruses Disease

- a general term used to describe infections caused by a group of viruses occur during warm weather months, when ticks are active
- Three clinical syndromes
 - Undifferentiated fever with or without maculopapular rash
 - Encephalitis
 - Incubation period is 4-21 days
 - Sudden onset of severe headache, chill and fever, nausea and vomiting, CNS depression, seizures, and mononuclear cells in cerebrospinal fluid , tremors convulsion and coma develops
 - Hemorrhagic fever
 - multiple hemorrhagic manifestations

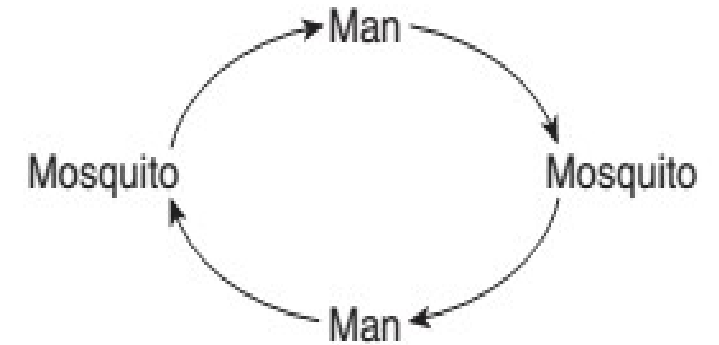
The zoonotic viruses are divided into two groups.

- **Arboviruses** those transmitted to humans by infected **bloodsucking insects** such as **mosquitoes, ticks, and *Phlebotomus* flies (sandflies)**.
 - **Other zoonotic RNA viruses** are transmitted by inhalation of infected animal excretions, by the conjunctival route, or occasionally by direct contact with infected animals.
- ❑ Rabies virus, which is commonly transmitted by animal bites
- In most cases, the **zoonotic viruses** were first named after the **place of initial isolation** (eg, St. Louis encephalitis) or
 - After the disease produced (eg, yellow fever)

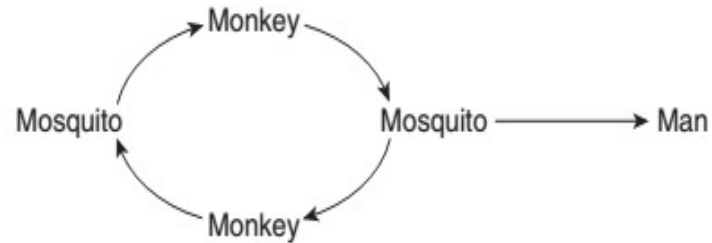
- Reservoirs are in nonhuman vertebrates
- Sometimes maintained by vertical transmission in vector
- Multiplication in vector is required

The three basic cycles of arbovirus transmission are:

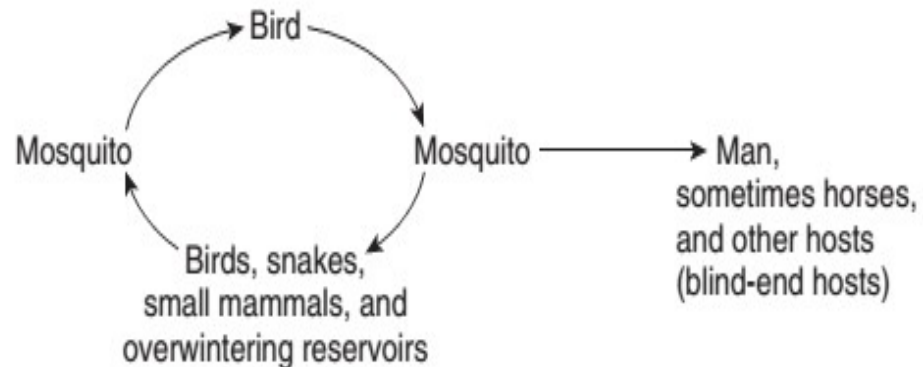
1. **Urban:** favored by the presence of large numbers of humans living in close proximity to arthropod species capable of virus transmission
- Urban cycle exists with **dengue and yellow fever**



2. **Sylvatic cycle** a single nonhuman vertebrate reservoir may be involved:

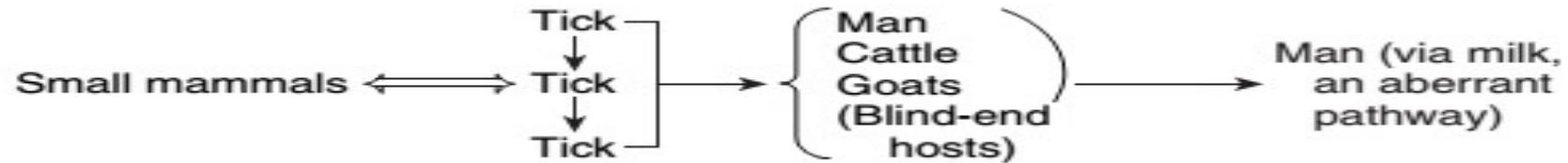


- Human are tangential host
- An example of this cycle is **jungle yellow fever**
- In other sylvatic cycles, multiple vertebrate reservoirs involved:
- Examples include **western equine encephalitis, eastern equine encephalitis, and California viruses.**



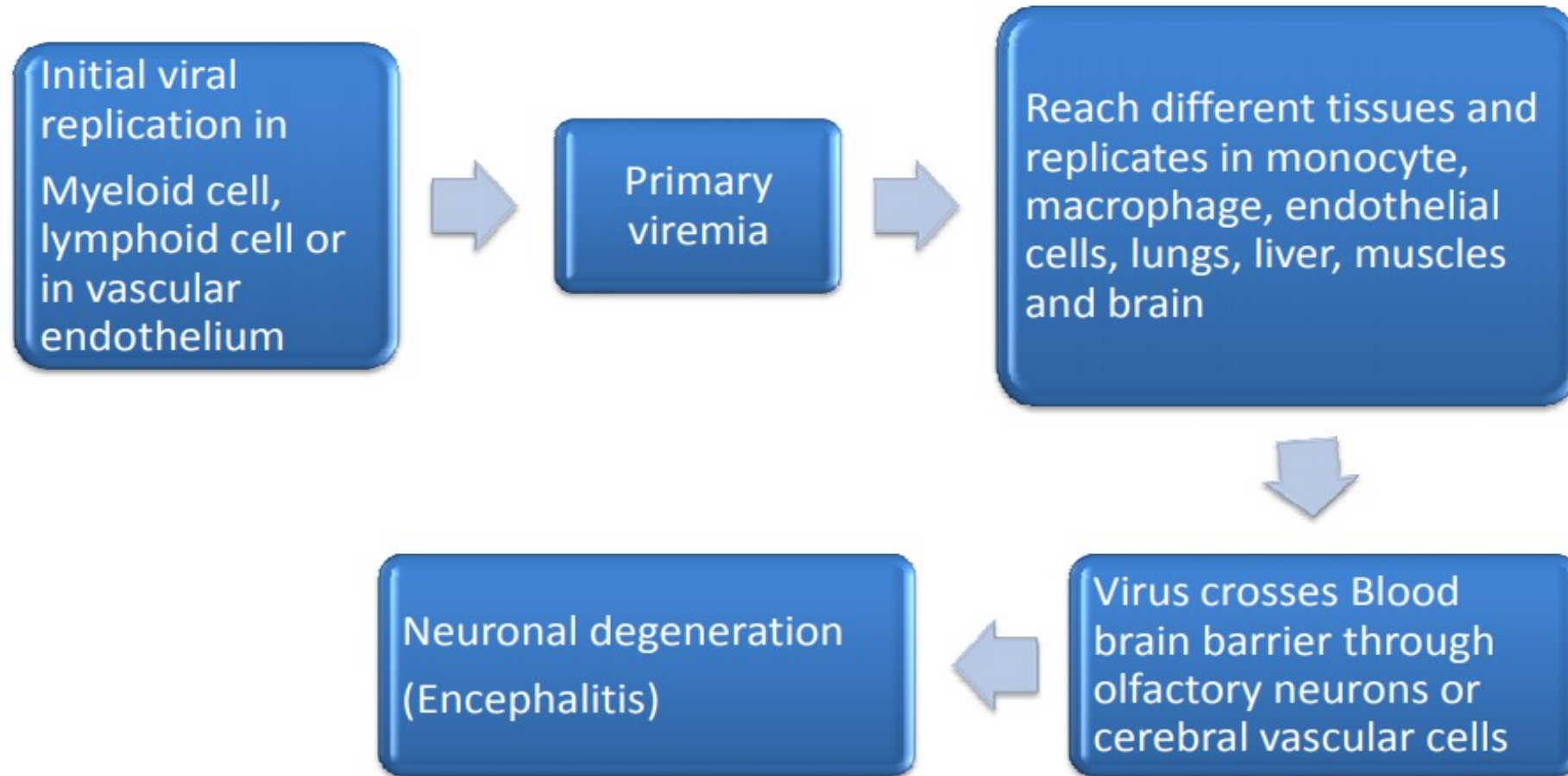
3. Arthropod-Sustained

- Arthropod sustained by tick **transovarial transmission**
- Weather, swamps, and ponds alter conditions



- Mosquito increases create risk for blind-end human infection
- Tick-borne encephalitis in Russia is transmitted by this cycle.
- An example of amplification is provided by **western equine encephalitis**

PATHOGENESIS AND PATHOLOGY OF ARBOVIRUS INFECTION



Diagnosis

- Blood is best source but must be early in disease
- Multiple serologic methods used

Treatment and prevention

- Treatment is only supportive
- Protection from bites and vector control are primary prevention
- Yellow fever vaccine is available

Rabies

- ✚ **Rabies** is an acute fatal viral illness of CNS.
- ✚ The word rabies is derived from the Latin verb to **rage**, which suggests the appearance of the **rabid patient**.
- ✚ It can affect all mammals
- ✚ It is transmitted between them by infected secretions, most often by bite.
- ✚ It was first recognized more than 3000 years ago and has been the most feared of infectious diseases.
- ✚ It is said that Aristotle recognized that rabies could be spread by a **rabid dog**.

Virology

- The rabies virus is a **bullet-shaped**, enveloped, ss RNA virus of the rhabdovirus group
- Rabies virus is large, with dimensions of about 180 by 70 nm.
- Knob-like glycoprotein, which elicit neutralizing and Hemagglutination inhibiting antibodies, cover the surface of the virion.
- Strains from different sources are antigenically heterogeneous

RABIES

- Rabies involves the development of **severe neurologic symptoms** and signs in a patient who was previously bitten by an animal.
- The neurologic abnormalities are very characteristic:
 - A relentlessly progressive excess of motor activity
 - Agitation
 - Hallucinations
 - Salivation
 - foaming at the mouth
 - severe throat contractions
- ❖ The **neurologic abnormalities** are explained by spread of the virus from the **bite wound** into the **CNS** and then **centrifugally** to the **autonomic nervous system**.

Pathogenesis

- Replicates initially in muscle and then enters peripheral nervous system
- Spreads to CNS gray matter
- The pathognomonic lesion is **Negri bodies** found in neurons
- Incubation period can be prolonged for months

Manifestations

- Encephalitis common, sometimes with ascending paralysis
- Almost uniformly fatal

Diagnosis

- Virus or antigen detected in brain tissue
- Histologic examination of their brain tissue shows **Negri bodies** in 80% of cases

Treatment

- No specific treatment is available

Prevention

- Vaccine-induced antibody inhibits viral spread
- High-risk individuals include veterinarians, spelunkers, laboratory workers require pre-exposure prophylaxis
 - Attenuated rabies virus grown in human diploid cell culture and inactivated with β -propiolactone.
 - Careful history and studies of biting animal are important in decision-making
 - Rabies immune globulin plus vaccine necessary in post-exposure management

Retroviruses

- Enveloped, spherical, + sensed ss RNA viruses
- Distinguished from all other RNA viruses by presence of an unusual enzyme, **reverse transcriptase**
- Representatives of two major genera:
 - **Oncoviruses** (*onco-*, “related to a tumor”)
 - Retinoblastoma (RB), p53, BRCA1, BRCA2
 - **Lentiviruses** (*lenti-*, “slow”)
 - HIV 1 & 2, Human T-cell leukemia virus (HTLV-1 and -2)
- and replicate through a DNA intermediate
- Many of these viruses **alter cellular growth** by expressing analogues of cellular growth-controlling genes

Human Immunodeficiency Virus

- The pandemic of AIDS (Acquired Immunodeficiency Disease Syndrome) has been with us for more than thirty years
 - about 39 million people have died of the disease
 - About 1.2 million people around the world die of AIDS each year.
 - Today, at least 36.9 million people are infected/ >7000 new infections every day
 - In 2015, 15 million infected persons were receiving anti-retroviral therapy
 - In sub-Saharan Africa, more than 22 million people are living with HIV infection
 - accounting for about 70% of total cases

HIV

- Genus lentivirus , a class of retrovirus, take a long time to cause overt disease, are the etiologic agents of AIDS
- Most lentiviruses target cells of the immune system and thus disease is often manifested as immunodeficiency.
- There are two types of HIV: HIV-1 and HIV-2
 - These cause clinically indistinguishable disease
 - The worldwide epidemic of HIV/AIDS by HIV-1
 - HIV-2 is mostly restricted to west Africa.
- The HIV are cytolytic and non-transforming
 - However, AIDS patients are at elevated risk of several types of cancer

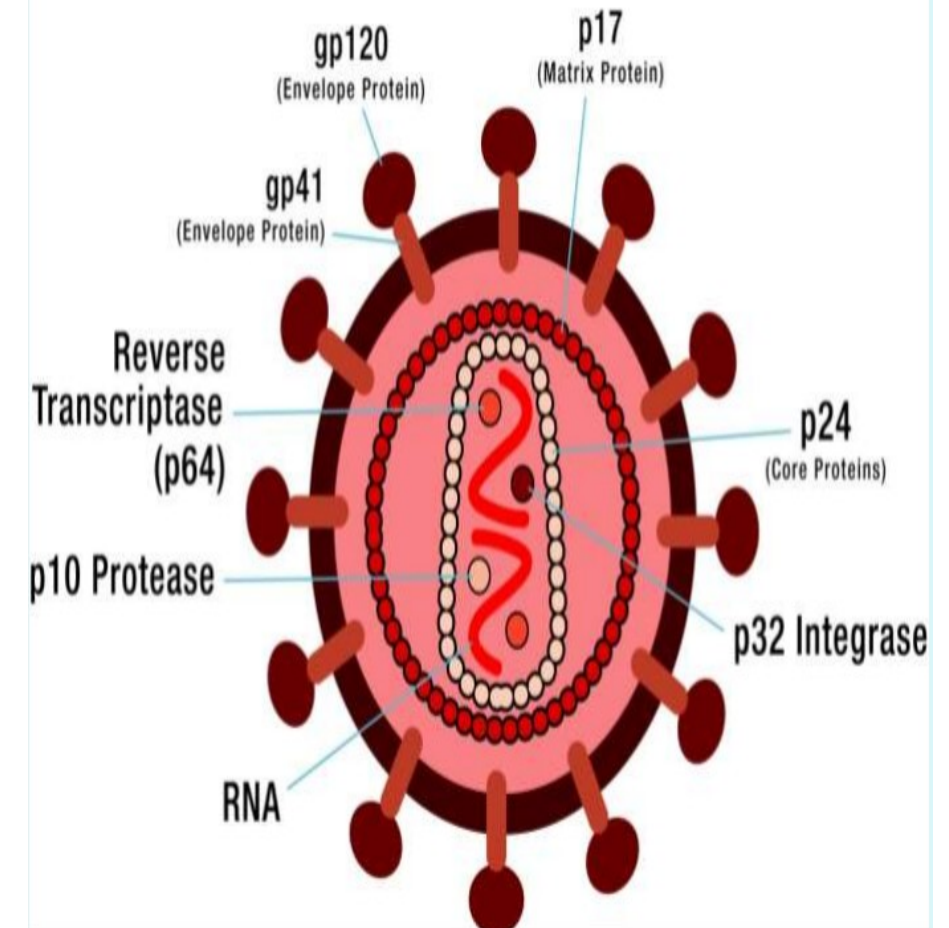
Structure and Composition

- HIVs are approximately 100 nm in diameter
- It has a **lipid envelope**,
 - embedded transmembrane glycoprotein **gp41** attached to the surface glycoprotein **gp120**
 - responsible for attachment to the host cell and are encoded by the **env gene** of the viral RNA genome.
- Beneath the envelope is the **matrix protein** p17, the core proteins p24 and p6 and the nucleocapsid protein p7
 - all encoded by the viral **gag gene**
- **Viral core**, lies 2 copies of positive-sense viral RNA genome with two cellular transfer RNA (tRNAs), protease, integrase, reverse transcriptase enzymes.
 - These three enzymes are encoded by the viral **pol gene**

Structure and Composition

Table 62-2 Retrovirus Genes and Their Function

Gene	Virus	Function
<i>gag</i>	All	Group-specific antigen: core and capsid proteins
<i>int</i>	All	Integrase
<i>pol</i>	All	Polymerase: reverse transcriptase, protease, integrase
<i>pro</i>	All	Protease
<i>env</i>	All	Envelope: glycoproteins
<i>tax</i>	HTLV	Transactivation of viral and cellular genes
<i>tat</i>	HIV-1	Transactivation of viral and cellular genes
<i>rex</i>	HTLV	Regulation of RNA splicing and promotion of export to cytoplasm
<i>rev</i>	HIV-1	Regulation of RNA splicing and promotion of export to cytoplasm
<i>nef</i>	HIV-1	Decreases cell surface CD4; facilitates T-cell activation; progression to AIDS (essential)
<i>vif</i>	HIV-1	Virus infectivity, promotion of assembly, blocks a cellular antiviral protein
<i>vpu</i>	HIV-1	Facilitates virion assembly and release, induces degradation of CD4
<i>vpr (vpx*)</i>	HIV-1	Transport of complementary DNA to nucleus, arresting of cell growth; replication in macrophages
LTR	All	Promoter, enhancer elements



Life cycle

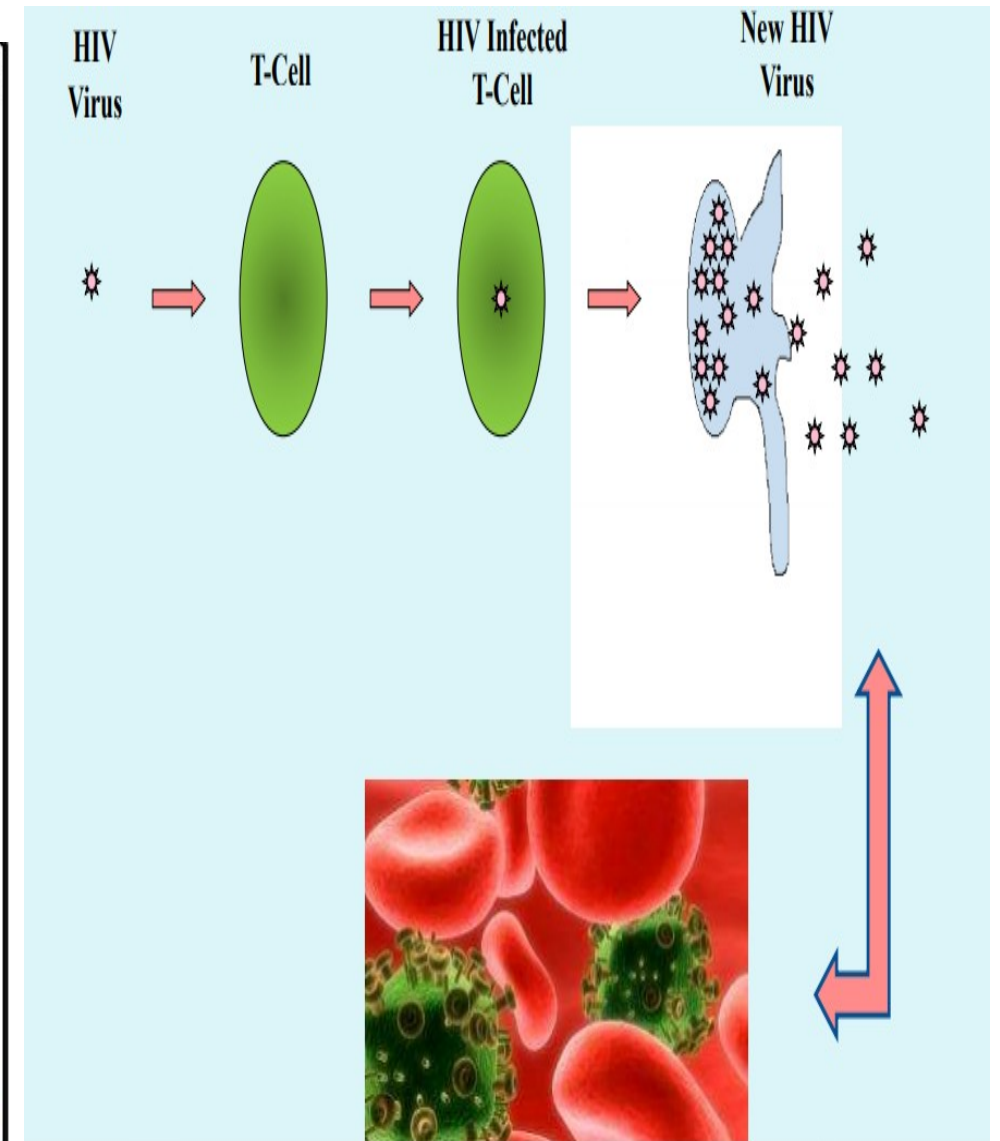
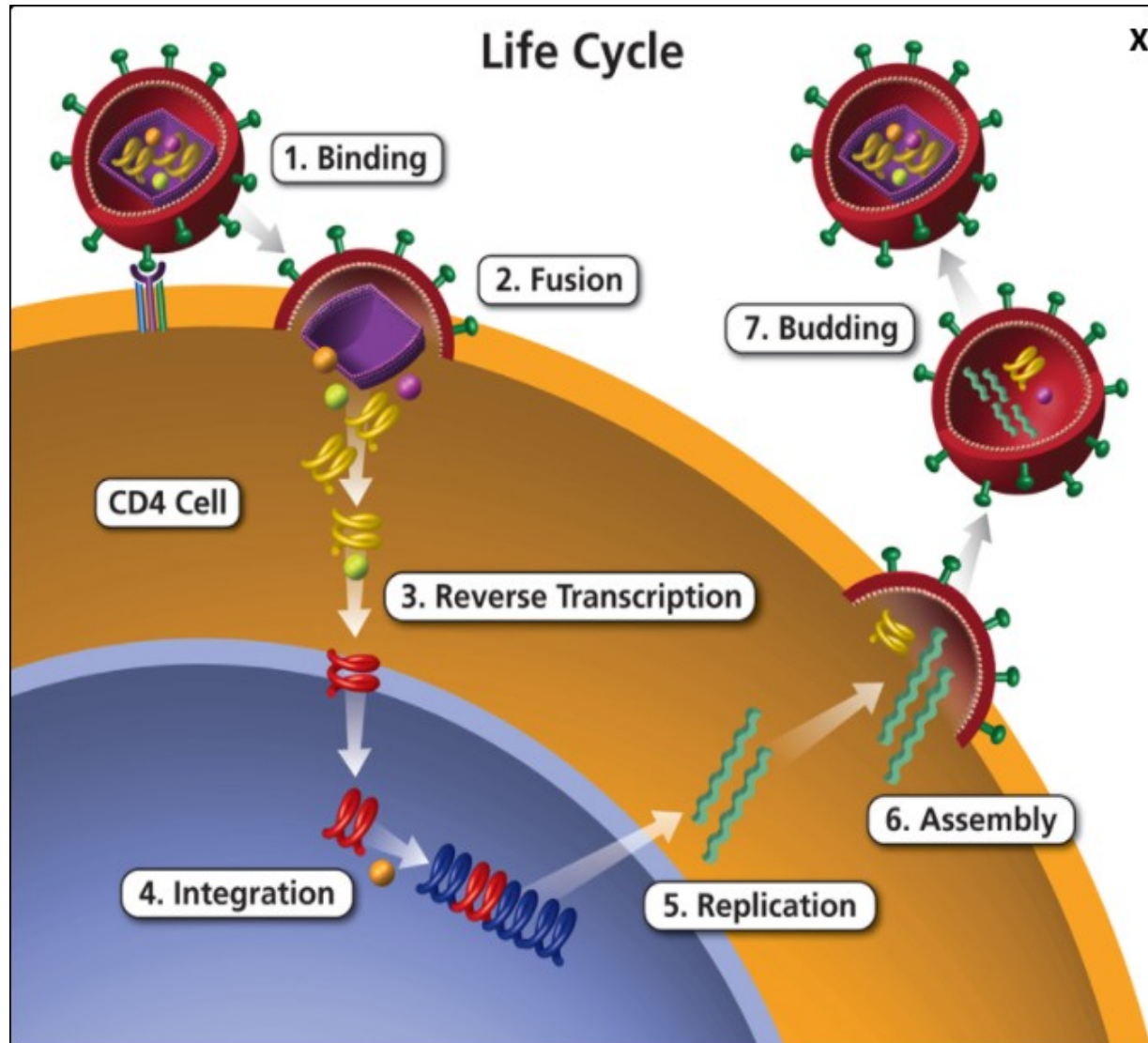
Viral Entry

- surface glycoprotein (gp120) attaches to CD4 cell and chemokine coreceptors, CXCR4 or CCR5
- **Transmembrane (TM) gp41** protein mediates fusion of viral and cell membranes
- Fusion provides direct cell-to-cell transmission

Viral RNA Replication

- Reverse transcriptase copies RNA to double-stranded DNA
- DNA integrates into host chromosome and replicates with the cell as a provirus
- Provirus includes its own promoter and signals that control transcription by host RNA polymerase
- Integrase-catalyzed integration is random in host DNA
- Integrated DNA is transcribed by host RNA polymerase

REPLICATION



AIDS

- Acquired immuno deficiency syndrome
- Fatal illness
- Caused by a retrovirus HIV
- It breaks down the body's immune system, leaving the patient vulnerable to a host of life threatening opportunistic infections, neurological disorders or unusual malignancies.

Epidemiology

- Males>females
- Occurs in all ages and ethnic groups
- AIDS is now the second leading cause of death for all men aged 25-44 years

Clinical Manifestations

Incubation Period (long)

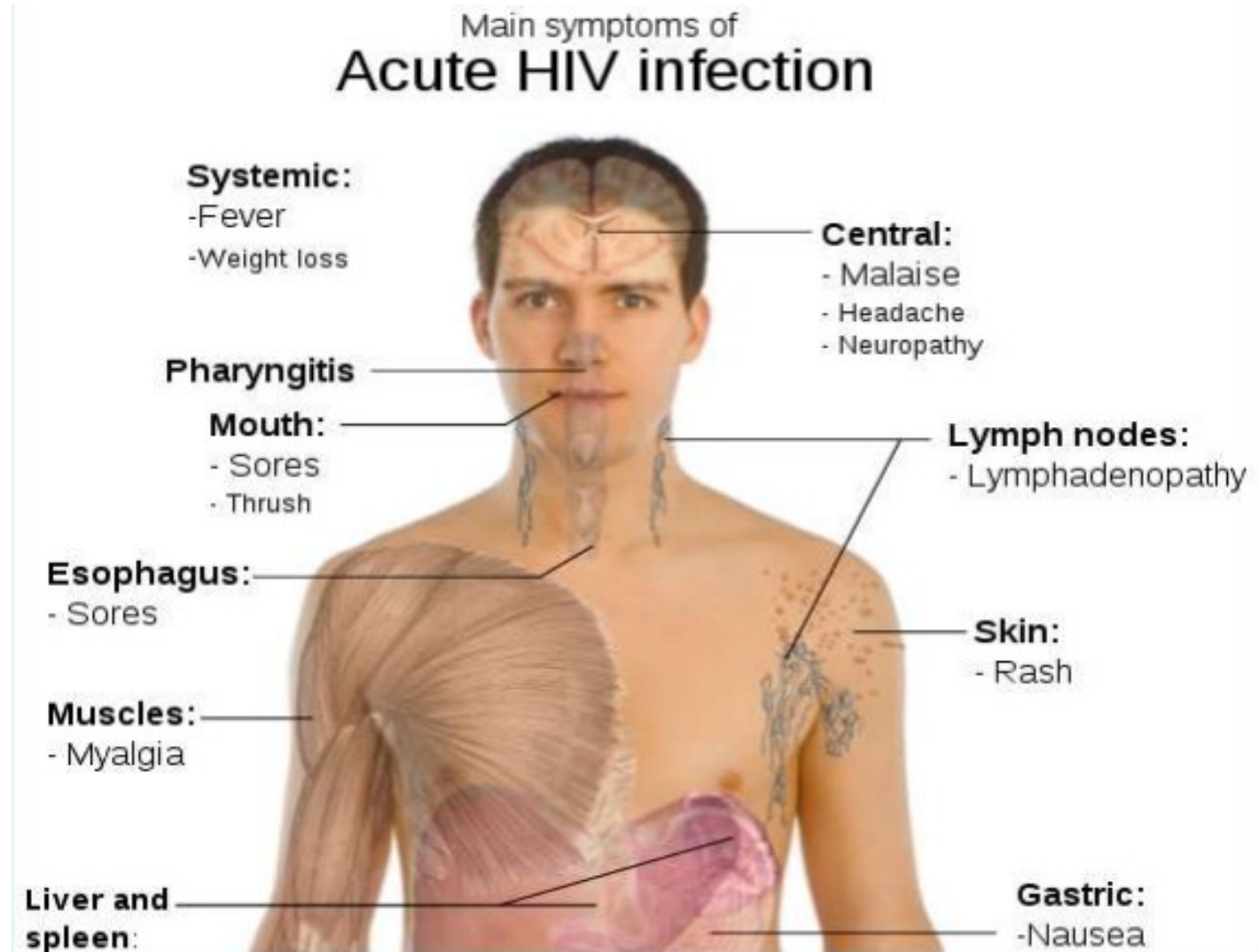
- Initial Infection
- Asymptomatic Carrier State
- AIDS-related Complex(ARC)
- AIDS

Initial Infection

- Except for a generally mild illness of fever,
 - sore throat and rash, which about 70% of the people experience a few weeks after the initial infection;
 - Most HIV – infected people have no symptoms for the first five years.
 - However they can infect others, Once, infected the infection is for life.
 - Antibody Response usually takes 2-12 weeks to appear in the blood stream. This period is called 'the window period'. (Tests- Negative)

The Acute HIV Syndrome

- Follows 3-6 wks following primary infection



Asymptomatic Carrier State

- Infected people with antibodies but without any overt signs of the disease,
 - persistent generalized lymphadenopathy.
 - viral replication occurring(up to 10 billion virions per day)
- It is however not firmly clear about how long does the asymptomatic stage lasts.

AIDS-Related Complex

- Has illnesses caused by damage to immune system, but without the opportunistic infections and cancers associated with AIDS.
- They may exhibit
 - Unexplained diarrhea(lasting more than a month), fatigue, malaise, loss of body weight(>10%), fever, night sweats.
 - CD4 counts drop to 500-600 cells/ml
 - Signs of Mild infections like oral thrush, generalized lymphadenopathy, enlarged spleen.

Common manifestation of AIDS

CD4 count less than 200/mm, opportunistic infections

- Lung infection:
 - ***Pneumocystis carinii*** pneumonia
- Gastrointestinal infection:
candidiasis of mouth or esophagus
- Skin infection:
 - Kaposi's sarcoma - red or violet macules or papules
- Central nervous System Infection:
 - Toxoplasmosis
 - Dementia
 - Meningitis
 - Primary CNS
 - Lymphomas.
 - Progressive Multifocal Leucoencephalopathy.



Transmission

- The HIV virus is transmitted between humans in three ways:
 - Sexually,
 - Perinatally, and
 - by exposure to contaminated blood or body fluids. average number of HIV particles in 1 ml of body fluids
 - Blood 18,000, Semen 11,000, Vaginal Fluid 7,000, Saliva 1
- Testing of blood supply reduced risk
- Intravenous drug abusers are at extremely high risk
- Accidental needle-sticks among health care workers mandate extreme care in prevention
- Shed in breast milk, where it may infect breastfeeding infants

Occurrence

- Prevalence rates have shifted over time, with increasing cases among **women** and economically disadvantaged minority groups
- Men and women nearly equally infected in Africa and Asia

Infection

- Major targets are CD4-bearing cells, but other cell types can also be infected
- Rapid turnover of CD4+cells during infection

Latency

- The long asymptomatic period following HIV infection (clinical latency) occurs despite active virus replication in the host.
- **Level of plasma viremia** directly correlates with disease progression

Immune Deficiency

- Immune deficiency related to reduction in numbers and normal functions of CD4+ T lymphocytes
- Infected individuals are susceptible to other infections and malignancies

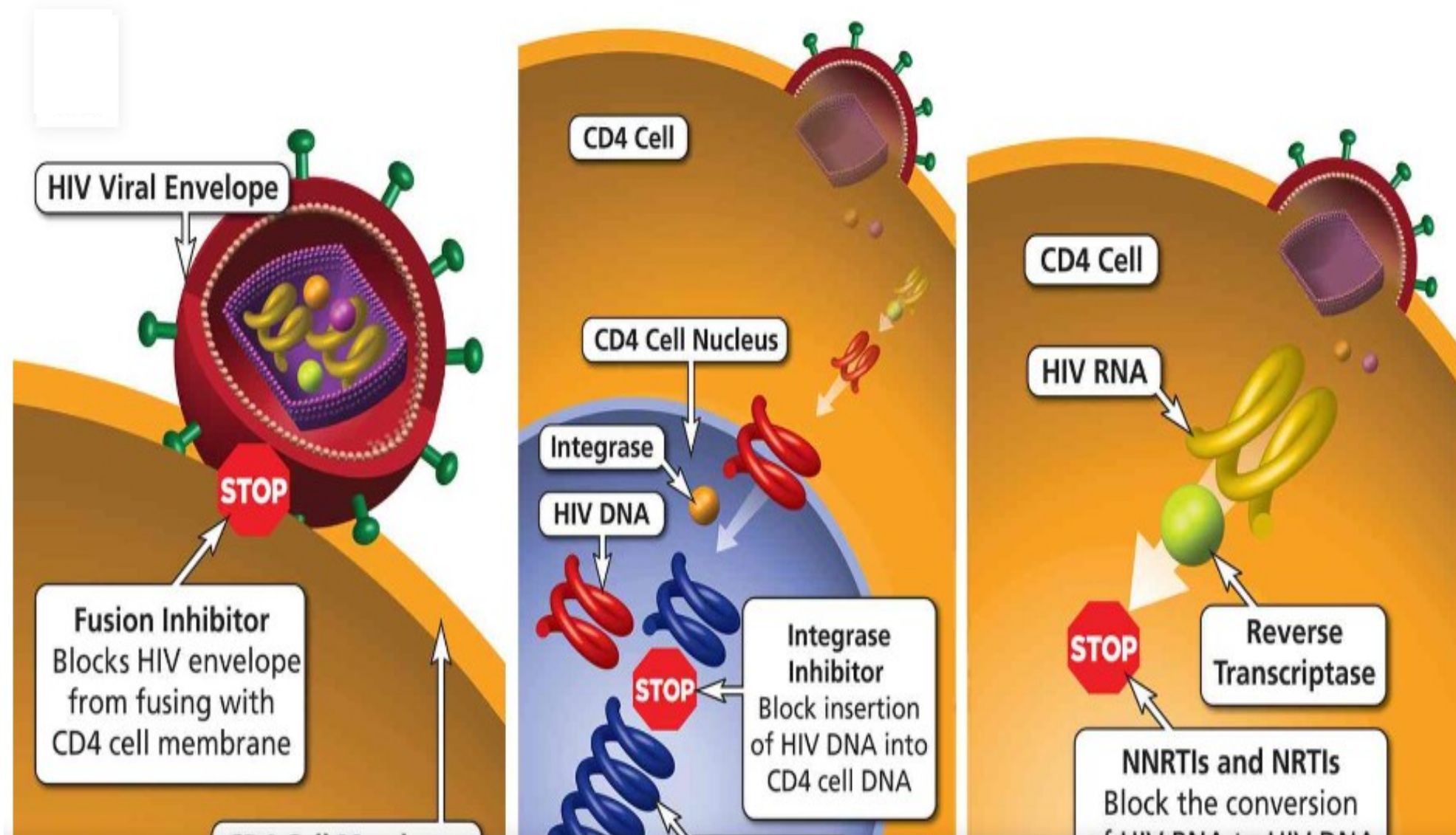
LABORATORY DIAGNOSIS

➤ Specific tests

- Antigen detection: p24 antigen
- Virus isolation
- Detection of viral nucleic acid
- Antibody detection

Treatment, Prevention, and Control

- A growing number of antiviral drugs are approved for treatment of HIV infections
- The anti-HIV drugs can be classified as
 - Binding,
 - fusion-penetration ,
 - nucleoside analogue reverse transcriptase inhibitors,
 - Non-nucleoside reverse transcriptase inhibitors,
 - integrase, or protease.
- Advent of highly active antiretroviral therapy (**HAART**) reduce progression of AIDS and the development of neurologic manifestations



Prevention

- Prophylaxis of opportunistic infections is especially important
- Education is the cornerstone of prevention
- Screening for asymptomatic infection in pregnancy aids effective prophylaxis